

ArXiv Scalar vs AMC Tensor-Setup Equations (Factorized Term-by-Term)

Left column: scalar factorized equations from `learn/axxiv_eq.tex` (both M-scheme and J-scheme factorized forms).
 Right column: AMC-generated equations from `amc/examples/sample_output` (when available). Beneath each AMC snippet we show the AMC input file used to generate it.

One-body I

M-scheme factorized

$$f_{ij}^I = \sum_{ab} (\chi_{ab}^\alpha \Gamma_{biaj} + \chi_{ab}^\alpha \Gamma_{aibj})$$

J-scheme factorized

$$f_{ij}^I = \delta_{j_i j_j} \hat{j}_i^{-2} \sum_{abJ} \hat{j}^2 \chi_{ab}^\alpha \Gamma_{biaj}^J$$

M-intermediate

$$\chi_{ij}^\alpha = \frac{1}{2} \sum_{abc} (\bar{n}_a \bar{n}_b n_c n_i - n_a n_b \bar{n}_c \bar{n}_i) \Omega_{ciab} \Omega_{abcj}$$

J-intermediate

$$\chi_{ij}^\alpha = \sum_{abcJ} \frac{\hat{j}^2}{2\hat{j}_i^2} (\bar{n}_a \bar{n}_b n_c n_i - n_a n_b \bar{n}_c \bar{n}_i) \Omega_{ciab}^J \Omega_{abcj}^J$$

One-body II

M-scheme factorized

$$f_{ij}^{II} = \sum_{ab} (\chi_{ab}^\beta \Omega_{biaj} - \chi_{ab}^\beta \Omega_{aibj})$$

J-scheme factorized

$$f_{ij}^{II} = \delta_{j_i j_j} \hat{j}_i^{-2} \sum_{abJ} \hat{j}^2 (\chi_{ab}^\beta - \chi_{ba}^\beta) \Omega_{biaj}^J$$

M-intermediate

$$\chi_{ij}^\beta = \frac{1}{2} \sum_{abc} (\bar{n}_a \bar{n}_b n_c n_i - n_a n_b \bar{n}_c \bar{n}_i) \Omega_{ciab} \Gamma_{abcj}$$

J-intermediate

$$\chi_{ij}^\beta = \sum_{abcJ} \frac{\hat{j}^2}{2\hat{j}_i^2} (\bar{n}_a \bar{n}_b n_c n_i - n_a n_b \bar{n}_c \bar{n}_i) \Omega_{ciab}^J \Gamma_{abcj}^J$$

AMC generated (tensor setup)

$$\chi_{ij}^{\alpha 0} = \frac{1}{2} \cdot \delta_{j_j, j_i} \cdot \hat{j}_i^{-2} \cdot \sum_{abcJ_0J_1\lambda_0} \cdot (\bar{n}_a \cdot \bar{n}_b \cdot n_c \cdot n_i - n_a \cdot n_b \cdot \bar{n}_c \cdot \bar{n}_i) \cdot (-1)^{J_0+J_1+\lambda_0} \cdot \lambda_0^{-1} \cdot \Omega_{ciab}^{J_0J_1\lambda_0} \cdot \Omega_{abcj}^{J_1J_0\lambda_0}$$

$$f_{ij}^{(I)0} =$$

$$\delta_{j_j, j_i} \cdot \hat{j}_i^{-2} \cdot \sum_{abJ_0} \cdot \delta_{j_b, j_a} \cdot \hat{j}_0^2 \cdot \chi_{ab}^{\alpha 0} \cdot \Gamma_{biaj}^{J_0J_0^0}$$

$$\delta_{j_j, j_i} \cdot \hat{j}_i^{-2} \cdot \sum_{abJ_0} \cdot \delta_{j_b, j_a} \cdot \hat{j}_0^2 \cdot \chi_{ab}^{\alpha 0} \cdot \Gamma_{aibj}^{J_0J_0^0}$$

AMC input: `amc/examples/sample_input/dc f1 tensor clean.txt`

AMC generated (tensor setup)

$$\chi_{ij}^{\beta \lambda_0} = \frac{1}{2} \cdot (-1)^{j_j+\lambda_0} \cdot \sum_{0abcJ_0J_1} \cdot (\bar{n}_a \cdot \bar{n}_b \cdot n_c \cdot n_i - n_a \cdot n_b \cdot \bar{n}_c \cdot \bar{n}_i) \cdot (-1)^{J_0+j_c} \cdot \hat{j}_0 \cdot \hat{j}_1 \cdot \left\{ \begin{matrix} J_0 & J_1 & \lambda_0 \\ j_j & j_i & j_c \end{matrix} \right\} \cdot \Omega_{ciab}^{J_0J_1\lambda_0} \cdot \Gamma_{abcj}^{J_1J_1^0}$$

AMC input: `amc/examples/sample_input/dc f2 chibeta tensor only.txt`
 (intermediate only)

One-body IIIa

M-scheme factorized

$$f_{ij}^{IIIa} = \sum_{abc} \left(\chi_{icab}^\gamma \Gamma_{abjc} - \Gamma_{ciab} \chi_{abcj}^\gamma \right)$$

J-scheme factorized

$$f_{ij}^{IIIa} = \delta_{j_i j_j} \hat{j}_i^{-2} \sum_{abcJ} \left(\bar{\chi}_{i\bar{c}a\bar{b}}^{\gamma J} \Gamma_{ab\bar{j}\bar{c}}^J - \bar{\chi}_{c\bar{j}a\bar{b}}^{\gamma J} \Gamma_{ab\bar{c}\bar{i}}^J \right)$$

M-intermediate

$$\chi_{ijkl}^\gamma = \sum_{ab} (n_a \bar{n}_b n_j \bar{n}_k - \bar{n}_a n_b \bar{n}_j n_k) \Omega_{ajkb} \Omega_{ibal}$$

J-intermediate

$$\bar{\chi}_{ijkl}^{\gamma J} = \sum_{ab} \hat{j}^2 (n_a \bar{n}_b \bar{n}_k n_l - \bar{n}_a n_b n_k \bar{n}_l) \bar{\Omega}_{ijab}^J \bar{\Omega}_{abk\bar{l}}^J$$

One-body IIIb

M-scheme factorized

$$f_{ij}^{IIIb} = \sum_{abc} \left(\chi_{ciab}^\delta \Gamma_{abcj} - \Gamma_{ciab} \chi_{abcj}^\delta \right)$$

J-scheme factorized

$$f_{ij}^{IIIb} = \delta_{j_i j_j} \hat{j}_i^{-2} \sum_{abcJ} \left(\chi_{ciab}^{\delta J} \Gamma_{abcj}^J - \chi_{abcj}^{\delta J} \Gamma_{ciab}^J \right)$$

M-intermediate

$$\chi_{ijkl}^\delta = \frac{1}{4} \sum_{ab} (\bar{n}_i \bar{n}_j n_a n_b - n_i n_j \bar{n}_a \bar{n}_b) \Omega_{ijab} \Omega_{abkl}$$

J-intermediate

$$\chi_{ijkl}^{\delta J} = \sum_{ab} \frac{\hat{j}^2}{4} (n_a n_b \bar{n}_k \bar{n}_l - \bar{n}_a \bar{n}_b n_k n_l) \Omega_{ijab}^J \Omega_{abkl}^J$$

AMC generated (tensor setup)

$$\begin{aligned} \chi_{ijkl}^{\gamma J_0 J_1 0} = & -\delta_{J_1, J_0} \cdot (-1)^{j_j + j_l} \cdot \sum_{abJ_2J_3J_4J_5J_6j_0\lambda_0} \\ & \cdot (n_a \cdot \bar{n}_b \cdot n_j \cdot \bar{n}_k - \bar{n}_a \cdot n_b \cdot \bar{n}_j \cdot n_k) \\ & \cdot (-1)^{J_3+J_5+\lambda_0} \cdot \hat{j}_2 \cdot \hat{j}_3 \cdot \hat{j}_4 \cdot \hat{j}_5 \cdot \hat{j}_6^2 \cdot \hat{j}_0^2 \\ & \cdot \lambda_0^{-1} \cdot \left\{ \begin{matrix} J_3 & \lambda_0 & J_2 \\ j_a & j_j & j_0 \end{matrix} \right\} \cdot \left\{ \begin{matrix} J_4 & \lambda_0 & J_5 \\ j_a & j_l & j_0 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_j & j_k & J_6 \\ j_b & j_0 & J_3 \end{matrix} \right\} \\ & \cdot \left\{ \begin{matrix} j_l & j_i & J_6 \\ j_b & j_0 & J_4 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_k & j_l & J_0 \\ j_i & j_j & J_6 \end{matrix} \right\} \cdot \Omega_{ajkb}^{J_2J_3\lambda_0} \cdot \Omega_{ibal}^{J_4J_5\lambda_0} \end{aligned}$$

$$f_{ij}^{(IIIa)0} =$$

$$\begin{aligned} & \delta_{j_j, j_i} \cdot \hat{j}_i^{-2} \cdot \sum_{abcJ_0} \cdot \hat{j}_0^2 \cdot \chi_{icab}^{\gamma J_0 J_0 0} \cdot \Gamma_{abcj}^{J_0 J_0 0} \\ & - \delta_{j_j, j_i} \cdot \hat{j}_i^{-2} \cdot \sum_{abcJ_0} \cdot \hat{j}_0^2 \cdot \Gamma_{ciab}^{J_0 J_0 0} \cdot \chi_{abcj}^{\gamma J_0 J_0 0} \end{aligned}$$

AMC input: `amc/examples/sample`input/dc`f3a`tensor`clean.txt`

AMC generated (tensor setup)

$$\begin{aligned} \chi_{ijkl}^{\delta J_0 J_1 0} = & \frac{1}{4} \cdot \delta_{J_1, J_0} \cdot (-1)^{J_0} \cdot \hat{j}_0^{-2} \cdot \sum_{abJ_2\lambda_0} \\ & \cdot (\bar{n}_i \cdot \bar{n}_j \cdot n_a \cdot n_b - n_i \cdot n_j \cdot \bar{n}_a \cdot \bar{n}_b) \\ & \cdot (-1)^{J_2+\lambda_0} \cdot \lambda_0^{-1} \cdot \Omega_{ijab}^{J_0 J_2 \lambda_0} \cdot \Omega_{abkl}^{J_2 J_0 \lambda_0} \end{aligned}$$

$$f_{ij}^{(IIIb)0} =$$

$$\begin{aligned} & \delta_{j_j, j_i} \cdot \hat{j}_i^{-2} \cdot \sum_{abcJ_0} \cdot \hat{j}_0^2 \cdot \chi_{ciab}^{\delta J_0 J_0 0} \cdot \Gamma_{abcj}^{J_0 J_0 0} \\ & - \delta_{j_j, j_i} \cdot \hat{j}_i^{-2} \cdot \sum_{abcJ_0} \cdot \hat{j}_0^2 \cdot \Gamma_{ciab}^{J_0 J_0 0} \cdot \chi_{abcj}^{\delta J_0 J_0 0} \end{aligned}$$

AMC input: `amc/examples/sample`input/dc`f3b`tensor`clean.txt`

Two-body I

M-scheme factorized

$$\Gamma_{ijkl}^I = \sum_a \left[(1 - \hat{P}_{ij}) \chi_{ai}^\epsilon \Gamma_{ajkl} + (1 - \hat{P}_{kl}) \chi_{ak}^\epsilon \Gamma_{ijal} \right]$$

J-scheme factorized

$$\Gamma_{ijkl}^{I,J} = \sum_a \left[(1 - \hat{P}_{ij}^J) \chi_{ai}^\epsilon \Gamma_{ajkl}^J + (1 - \hat{P}_{kl}^J) \chi_{ak}^\epsilon \Gamma_{ijal}^J \right]$$

M-intermediate

$$\chi_{ij}^\epsilon = \frac{1}{2} \sum_{abc} (\bar{n}_a \bar{n}_b n_c + n_a n_b \bar{n}_c) \Omega_{ciab} \Omega_{abcj}$$

J-intermediate

$$\chi_{ij}^\epsilon = \frac{1}{2 \hat{j}_j^2} \sum_{abcJ} \hat{j}^2 (\bar{n}_a \bar{n}_b n_c + n_a n_b \bar{n}_c) \Omega_{ciab}^J \Omega_{abcj}^J$$

Two-body II

M-scheme factorized

$$\Gamma_{ijkl}^{II} = \sum_a \left[(1 - \hat{P}_{ij}) \chi_{aj}^\zeta \Omega_{iakl} - (1 - \hat{P}_{kl}) \chi_{ak}^\zeta \Omega_{ijal} \right]$$

J-scheme factorized

$$\Gamma_{ijkl}^{II,J} = \sum_a \left[(1 - \hat{P}_{ij}^J) \chi_{aj}^\zeta \Omega_{iakl}^J - (1 - \hat{P}_{kl}^J) \chi_{ak}^\zeta \Omega_{ijal}^J \right]$$

M-intermediate

$$\chi_{ij}^\zeta = \frac{1}{2} \sum_{abc} (n_a n_b \bar{n}_c + \bar{n}_a \bar{n}_b n_c) \Gamma_{aibc} \Omega_{bcaj}$$

J-intermediate

$$\chi_{ij}^\zeta = \frac{1}{2} \sum_{abcJ} \hat{j}^2 \hat{j}_j^{-2} (\bar{n}_a \bar{n}_b n_c + n_a n_b \bar{n}_c) \Gamma_{aibc}^J \Omega_{bcaj}^J$$

AMC generated (tensor setup)

$$\chi_{ij}^{\epsilon \ 0} = \frac{1}{2} \cdot \delta_{jj,ji} \cdot \hat{j}_i^{-2} \cdot \sum_{abcJ_0J_1\lambda_0} \cdot (\bar{n}_a \cdot \bar{n}_b \cdot n_c + n_a \cdot n_b \cdot \bar{n}_c) \cdot (-1)^{J_0+J_1+\lambda_0} \cdot \lambda_0^{-1} \cdot \Omega_{ciab}^{J_0J_1\lambda_0} \cdot \Omega_{abcj}^{J_1J_0\lambda_0}$$

$$\Gamma_{ijkl}^{(I) \ J_0J_10} = \delta_{J_1,J_0} \cdot \left[(1 - \hat{P}_{ij}) \sum_a \delta_{ja,ji} \cdot \chi_{ia}^{\epsilon \ 0} \cdot \Gamma_{ajkl}^{J_0J_00} + (1 - \hat{P}_{kl}) \sum_a \delta_{ja,jk} \cdot \chi_{ak}^{\epsilon \ 0} \cdot \Gamma_{ijal}^{J_0J_00} \right]$$

AMC input: `amc/examples/sample`input/dc`G1`chiepsilon`core`only`

AMC generated (tensor setup)

$$\chi_{ij}^{\beta \ \lambda_0} = \frac{1}{2} \cdot (-1)^{j_j+\lambda_0} \cdot \sum_{0abcJ_0J_1} \cdot (\bar{n}_a \cdot \bar{n}_b \cdot n_c \cdot n_i - n_a \cdot n_b \cdot \bar{n}_c \cdot \bar{n}_i) \cdot (-1)^{J_0+j_c} \cdot \hat{J}_0 \cdot \hat{J}_1 \cdot \left\{ \begin{matrix} J_0 & J_1 & \lambda_0 \\ j_j & j_i & j_c \end{matrix} \right\} \cdot \Omega_{ciab}^{J_0J_1\lambda_0} \cdot \Gamma_{abcj}^{J_1J_10}$$

AMC (intermediate):
`amc/examples/sample`input/dc`f2`chibeta`tensor`only.txt`
 AMC (closure):
`amc/examples/sample`input/dc`f2`chibeta`closure`only.txt`

Two-body IIIa

M-scheme factorized

$$\Gamma_{ijkl}^{IIIa} = - \sum_{ab} \left[(1 - \hat{P}_{ij}) \chi_{ijab}^\eta \Gamma_{abkl} + (1 - \hat{P}_{kl}) \Gamma_{ijab} \chi_{abkl}^\eta \right]$$

J-scheme factorized

$$\Gamma_{ijkl}^{IIIa,J} = \sum_{ab} \left[(1 - \hat{P}_{ij}^J) \chi_{ijab}^{\eta J} \Gamma_{abkl}^J + (1 - \hat{P}_{kl}^J) \Gamma_{ijab}^J \chi_{abkl}^{\eta J} \right]$$

M-intermediate

$$\chi_{ijkl}^\eta = \sum_{ab} (\bar{n}_a n_b \bar{n}_k + n_a \bar{n}_b n_k) \Omega_{iabl} \Omega_{bjka}$$

J-intermediate

$$\bar{\chi}_{ij\bar{k}\bar{l}}^{\eta J} = \sum_{ab} (\bar{n}_a n_b \bar{n}_k + n_a \bar{n}_b n_k) \bar{\Omega}_{ij\bar{a}\bar{b}}^J \bar{\Omega}_{a\bar{b}\bar{k}\bar{l}}^J$$

Two-body IIIb

M-scheme factorized

$$\Gamma_{ijkl}^{IIIb} = -(1 - \hat{P}_{ij})(1 - \hat{P}_{kl}) \sum_{ab} \left(\chi_{bkai}^\eta \Gamma_{jbla} + \chi_{lajb}^\eta \Gamma_{aibk} \right)$$

J-scheme factorized

$$\bar{\Gamma}_{j\bar{l}k\bar{i}}^{IIIb,J} = (1 - \hat{P}_{ij}^J)(1 - \hat{P}_{kl}^J) \sum_{ab} \bar{\Gamma}_{j\bar{l}a\bar{b}}^J \left(\bar{\chi}_{a\bar{b}k\bar{i}}^J + \bar{\chi}_{k\bar{i}b\bar{a}}^J \right)$$

M-intermediate

$$\chi_{ijkl}^\eta = \sum_{ab} (\bar{n}_a n_b \bar{n}_k + n_a \bar{n}_b n_k) \Omega_{iabl} \Omega_{bjka}$$

J-intermediate

$$\bar{\chi}_{ij\bar{k}\bar{l}}^{\eta J} = \sum_{ab} (\bar{n}_a n_b \bar{n}_k + n_a \bar{n}_b n_k) \bar{\Omega}_{ij\bar{a}\bar{b}}^J \bar{\Omega}_{a\bar{b}\bar{k}\bar{l}}^J$$

AMC generated (tensor setup)

$$\begin{aligned} \chi_{ijkl}^{\eta J_0 J_1 0} = & -\delta_{J_1, J_0} \cdot (-1)^{j_i + j_k} \cdot \sum_{ab J_2 J_3 J_4 J_5 J_6 j_0 \lambda_0} \\ & \cdot (\bar{n}_a \cdot n_b \cdot \bar{n}_k + n_a \cdot \bar{n}_b \cdot n_k) \cdot (-1)^{J_2 + J_4 + \lambda_0} \\ & \cdot \hat{J}_2 \cdot \hat{J}_3 \cdot \hat{J}_4 \cdot \hat{J}_5 \cdot \hat{J}_6^2 \cdot \hat{j}_0^2 \cdot \lambda_0^{\hat{1}} \cdot \left\{ \begin{matrix} J_3 & \lambda_0 & J_2 \\ j_a & j_i & j_0 \end{matrix} \right\} \\ & \cdot \left\{ \begin{matrix} J_4 & \lambda_0 & J_5 \\ j_a & j_k & j_0 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_i & j_l & J_6 \\ j_b & j_0 & J_3 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_k & j_j & J_6 \\ j_b & j_0 & J_4 \end{matrix} \right\} \\ & \cdot \left\{ \begin{matrix} j_l & j_k & J_0 \\ j_j & j_i & J_6 \end{matrix} \right\} \cdot \Omega_{iabl}^{J_2 J_3 \lambda_0} \cdot \Omega_{bjka}^{J_4 J_5 \lambda_0} \end{aligned}$$

$$\begin{aligned} \Gamma_{ijkl}^{(IIIa) J_0 J_1 0} = & \delta_{J_1, J_0} \cdot \left[(1 - \hat{P}_{ij}) \sum_{ab} \cdot \chi_{ijab}^{\eta J_0 J_0 0} \cdot \Gamma_{abkl}^{J_0 J_0 0} \right. \\ & \left. + (1 - \hat{P}_{kl}) \sum_{ab} \cdot \Gamma_{ijab}^{J_0 J_0 0} \cdot \chi_{abkl}^{\eta J_0 J_0 0} \right] \end{aligned}$$

AMC *input* (core *kernels*):
amc/examples/sample`input/dc`G3a`chieta`core`only.txt

AMC generated (tensor setup)

$$\begin{aligned} \chi_{ijkl}^{\eta J_0 J_1 0} = & -\delta_{J_1, J_0} \cdot (-1)^{j_i + j_k} \cdot \sum_{ab J_2 J_3 J_4 J_5 J_6 j_0 \lambda_0} \\ & \cdot (\bar{n}_a \cdot n_b \cdot \bar{n}_k + n_a \cdot \bar{n}_b \cdot n_k) \cdot (-1)^{J_2 + J_4 + \lambda_0} \\ & \cdot \hat{J}_2 \cdot \hat{J}_3 \cdot \hat{J}_4 \cdot \hat{J}_5 \cdot \hat{J}_6^2 \cdot \hat{j}_0^2 \cdot \lambda_0^{\hat{1}} \cdot \left\{ \begin{matrix} J_3 & \lambda_0 & J_2 \\ j_a & j_i & j_0 \end{matrix} \right\} \\ & \cdot \left\{ \begin{matrix} J_4 & \lambda_0 & J_5 \\ j_a & j_k & j_0 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_i & j_l & J_6 \\ j_b & j_0 & J_3 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_k & j_j & J_6 \\ j_b & j_0 & J_4 \end{matrix} \right\} \\ & \cdot \left\{ \begin{matrix} j_l & j_k & J_0 \\ j_j & j_i & J_6 \end{matrix} \right\} \cdot \Omega_{iabl}^{J_2 J_3 \lambda_0} \cdot \Omega_{bjka}^{J_4 J_5 \lambda_0} \end{aligned}$$

$$\begin{aligned} \Gamma_{ijkl}^{(IIIb) J_0 J_1 0} = & (1 - \hat{P}_{ij})(1 - \hat{P}_{kl}) \cdot \left[-\delta_{J_1, J_0} \right. \\ & \cdot (-1)^{J_0 + j_i + j_j + j_k + j_l} \cdot \sum_{ab J_2 J_3} \cdot (-1)^{J_2 + J_3 + j_a + j_b} \\ & \cdot \hat{J}_2^2 \cdot \hat{J}_3^2 \cdot \left\{ \begin{matrix} j_k & j_b & J_2 \\ j_l & J_3 & j_a \\ J_0 & j_j & j_i \end{matrix} \right\} \cdot \chi_{bkai}^{\eta J_2 J_2 0} \\ & \cdot \Gamma_{jbla}^{J_3 J_3 0} - \delta_{J_1, J_0} \\ & \cdot (-1)^{J_0 + j_i + j_j + j_k + j_l} \cdot \sum_{ab J_2 J_3} \cdot (-1)^{J_2 + J_3 + j_a + j_b} \\ & \cdot \hat{J}_2^2 \cdot \hat{J}_3^2 \cdot \left\{ \begin{matrix} j_l & j_a & J_2 \\ j_k & J_3 & j_b \\ J_0 & j_i & j_j \end{matrix} \right\} \cdot \chi_{lajb}^{\eta J_2 J_2 0} \cdot \Gamma_{aibk}^{J_3 J_3 0} \left. \right] \end{aligned}$$

AMC *input* (core *kernels*):
amc/examples/sample`input/dc`G3b`chieta`core`only.txt

Two-body IIc

M-scheme factorized

$$\Gamma_{ijkl}^{III_c} = -\frac{1}{2}(1 - \hat{P}_{ij})(1 - \hat{P}_{kl}) \sum_{ab} \chi_{iabl}^{\theta} \Gamma_{bjka}$$

J-scheme factorized

$$\bar{\Gamma}_{i\bar{l}k\bar{j}}^{III_c,J} = \frac{1}{2}(1 - \hat{P}_{ij}^J)(1 - \hat{P}_{kl}^J) \sum_{ab} \bar{\chi}_{i\bar{l}a\bar{b}}^{\theta,J} \bar{\Gamma}_{a\bar{b}k\bar{j}}^J$$

M-intermediate

$$\chi_{ijkl}^\theta = \sum_{ab} (n_a n_b \bar{n}_k + \bar{n}_a \bar{n}_b n_k + n_a n_b \bar{n}_j + \bar{n}_a \bar{n}_b n_j) \Omega_{ijab} \Omega_{abkl}$$

J-intermediate

$$\chi_{ijkl}^{\theta J} = \sum_{ab} (n_a n_b \bar{n}_k + \bar{n}_a \bar{n}_b n_k + n_a n_b \bar{n}_j + \bar{n}_a \bar{n}_b n_j) \Omega_{ijab}^J \Omega_{abkl}^J$$

Two-body IVa

M-scheme factorized

$$\Gamma_{ijkl}^{IV_a} = - \sum_{ab} \left[(1 - \hat{P}_{ij}) \chi_{ijab}^{\kappa} \Omega_{bakl} - (1 - \hat{P}_{kl}) \Omega_{ijab} \chi_{klba}^{\kappa} \right]$$

J-scheme factorized

$$\Gamma_{ijkl}^{IV_{a,J}} = - \sum_{ab} \left[(1 - \hat{P}_{ij}^J) \chi_{ijab}^{\kappa J} \Omega_{abkl}^J + (1 - \hat{P}_{kl}^J) \Omega_{ijab}^J \chi_{klab}^{\kappa J} \right]$$

M-intermediate

$$\chi_{ijkl}^{\kappa} = \sum_{ab} (\bar{n}_a n_b n_k + n_a \bar{n}_b \bar{n}_k) \Omega_{ajbl} \Gamma_{ibka}$$

J-intermediate

$$\bar{\bar{\chi}}_{ij\bar{k}\bar{l}}^{\kappa J} = \sum_{ab} (\bar{n}_a n_b n_l + n_a \bar{n}_b \bar{n}_l) \bar{\bar{\Omega}}_{ij\bar{b}\bar{a}}^J \bar{\bar{\Gamma}}_{\bar{b}\bar{a}k\bar{l}}^J$$

Two-body IVb

M-scheme factorized

$$\Gamma_{ijkl}^{IV_b} = (1 - \hat{P}_{ij})(1 - \hat{P}_{kl}) \sum_{ab} (\chi_{aibk}^\iota \Omega_{jb la} - \chi_{akbi}^\iota \Omega_{ja lb})$$

J-scheme factorized

$$\overline{\overline{\Gamma}}_{j\bar{l}k\bar{i}}^{IV,J} = (1 - \hat{P}_{ij}^J)(1 - \hat{P}_{kl}^J) \sum_{ab} \overline{\overline{\Omega}}_{j\bar{l}a\bar{b}}^J (\bar{\chi}_{abk\bar{i}}^{IJ} - \bar{\chi}_{k\bar{i}a\bar{b}}^{IJ})$$

M-intermediate

$$\chi_{ijkl}^\iota = \sum_{ab} (\bar{n}_a n_b \bar{n}_k + n_a \bar{n}_b n_k) \Omega_{bika} \Gamma_{iabl}$$

J-intermediate

$$\bar{\chi}_{ij\bar{k}\bar{l}}^J = \sum_{ab} (\bar{n}_a n_b \bar{n}_k + n_a \bar{n}_b n_k) \bar{\Gamma}_{ij\bar{a}\bar{b}}^J \bar{\Omega}_{a\bar{b}k\bar{l}}^J$$

AMC generated (tensor setup)

$$\begin{aligned} \chi_{ijkl}^{\theta \quad J_0 J_1 0} &= \delta_{J_1, J_0} \cdot (-1)^{J_0} \cdot J_0^{\wedge 2} \cdot \sum_{ab J_2 \lambda_0} \\ &\quad \cdot (n_a \cdot n_b \cdot \bar{n}_k + \bar{n}_a \cdot \bar{n}_b \cdot n_k + n_a \cdot n_b \cdot \bar{n}_j + \bar{n}_a \cdot \bar{n}_b \cdot n_j) \\ &\quad \cdot (-1)^{J_2 + \lambda_0} \cdot \lambda_0^{\wedge 1} \cdot \Omega_{ijab}^{J_0 J_2 \lambda_0} \cdot \Omega_{abkl}^{J_2 J_0 \lambda_0} \end{aligned}$$

$$\Gamma_{ijkl}^{(III_c)J_0J_1J_0} = (1 - \hat{P}_{ij})(1 - \hat{P}_{kl}) \cdot \left[-\frac{1}{2} \cdot \delta_{J_1, J_0} \cdot \sum_{abJ_2J_3} \cdot \hat{J}_2^2 \right. \\ \left. \cdot \hat{J}_3^2 \cdot \begin{Bmatrix} j_i & j_a & J_2 \\ j_j & J_3 & j_b \\ J_0 & j_k & j_l \end{Bmatrix} \cdot \chi_{iabl}^\theta \cdot \Gamma_{bjka}^{J_2J_0J_3} \right]$$

AMC input (core kernels):
 amc/examples/sample_input/dc_G3c_chitheta_core_only.txt

AMC generated (tensor setup)

*Attempted AMC inputs: amc/examples/sample`input/dc`G4a`chikap`
(intermediate), amc/examples/sample`input/dc`G4a`chikappa`closur`
(closure attempt)*

Status: no matching factorized AMC closure available

AMC generated (tensor setup)

Attempted AMC inputs: `amc/examples/sample`input/dc`G4b`chiiota``
(intermediate), `amc/examples/sample`input/dc`G4b`chiiota`closure``
(closure attempt)

Status: no matching factorized AMC closure available